

SRI HARSHA

SOFTWARE ENGINEERING ARCHITECT

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20+ Years Experience
US Patent Holder
3B Events/Hour Processed
3× Products Taken 0 to Market

PROFESSIONAL SUMMARY

- ▶ Highly accomplished professional with 20+ years in product development, specializing in telecommunications and AI/ML.
- ▶ Currently driving strategic adoption of AI applications and Agentic technologies within the organization.
- ▶ Experience building full-stack AI/LLM/ML infrastructure and application solutions with AIOps approach.
- ▶ Expert in adopting open-source into enterprise solutions — accelerating development and achieving cost optimization.
- ▶ Extensive expertise in cloud-native, microservices-based systems architecture and delivery.
- ▶ Pioneered CI/CD, observability platforms, and test automation frameworks well before industry standardization.

ENGINEERING EXPERIENCE — TWO ERAS

PRE-AGENTIC ERA Deep Systems & Telecom Engineering (2004–2021)	AGENTIC ERA AI/LLM Architect & Agentic Development (2021–Present)
▶ 20+ years deep expertise in C/C++, Golang, Python across telecom and systems programming ▶ Engineered S2a (SaMOG) Wi-Fi Offload gateway features at Cisco (2015-2021) ▶ Implemented 5G-NSA solutions on MME/PGW — world's first 5G-NSA at T-Mobile ▶ Led 5G SMF QoS Modeling, SMF-UPF, and SMF-CHF communications at Cisco ▶ Architected world's first commercial 3GPP Femto Gateway (HNbGW) for SFR, France at Nokia ▶ FCAPS management for 60,000 concurrent Femto APs with 99.999% availability ▶ Led Agile transformation for 120-member cross-functional team at Nokia ▶ Pioneered CI/CD before Jenkins: 5,000+ test cases in 3 hours (Perl/Expect, 2013) ▶ TAC leadership for Jio — world's first greenfield all-IP 4G/LTE network at Samsung ▶ Managed KPIs for 100,000+ eNodeBs ▶ Feature Database and OTA updates for Motorola P2K Platform (2004-2008) ▶ Designed systems scaling 6X customers via automation at Motorola ▶ US Patent: US011350319B2 — Optimized QoS Enforcement for Handover Between Radio Access Types ▶ Expertise in P4 programming (Intel Tofino) processing Terabits/second ▶ Built MPTCP prototype porting TCP stack to DPDK for ultra-high throughput ▶ Architected Ribbon Analytics processing 3B events/hour serving 2.5M customers at Airtel India ▶ Full 5G stack analytics: AMF, AUSF, SMF, gNodeB, UPF, PCF, UDM, CHF, NRF, NSSF, SCP	▶ Spec-Based & Test-Driven Development: Architect-grade FSD/DSD → Claude Code/Copilot/Codex pipelines ▶ Rich experience authoring PRDs, FSDs, DSDs — now paired with AI coding agents for 10x delivery speed ▶ Pragmatic orchestration: spec → agentic stub → TDD loops with GitHub Copilot, Windsurf, Cursor, Claude Code ▶ LLM Application Development: RAG, document classification, sentiment analysis, fine-tuning, prompt engineering ▶ On-premises LLM deployments: NVIDIA M60/A100/H100/L40S GPUs — data sovereignty at enterprise scale ▶ RFx Responder: agentic AI reduced sales first response from months to days; 75% → 87% accuracy ▶ RAG Chatbot: 100+ GB data, 10M+ tokens served, \$100K+ annual savings for 1,000s of users ▶ Fine-tuning pipelines with Unsloth, LoRA, QLoRA, RLHF, RLAI — custom LLMs and SLMs ▶ Acumen AI Cluster: 5-node K8s, 12 GPUs (8×L40S + 4×Tesla M60), 230 GB VRAM, 192 virtual GPU slices ▶ Acumen AI Platform: self-service FastAPI + Celery + Flask + MCP Servers on distributed K8s ▶ Langflow/n8n agentic workflow builder — democratizing AI for non-developers ▶ AI/ML Steering Committee member driving enterprise-wide GenAI roadmap and adoption ▶ Introduced and drove org-wide adoption of Codeium, Windsurf, Cursor ▶ Moderated internal AI/ML forum accelerating GenAI adoption across org ▶ Scaled development/test optimization using agentic coding tools across multiple teams ▶ LLM Observability: Arize-Phoenix, Langfuse — tracing, token tracking, active feedback loop for fine-tuning ▶ eBPF kernel-level monitoring integration for enhanced network product capabilities

The Agentic Advantage

As an architect with hands-on experience writing Feature Specification Documents (FSDs) and Design Specification Documents (DSDs), I bring a unique edge: I translate rigorous, spec-based engineering directly into agentic workflows. Pairing architect-grade PRD/FSD authoring with Claude Code, GitHub Copilot, and Codex — and enforcing Test-Driven Development throughout — I deliver production-grade systems at dramatically accelerated timelines. The depth of my pre-agentic telecom and systems engineering grounds every AI-assisted decision in real-world architecture principles.

WORK PROFILE

Ribbon Communications | System Architecture Architect, Ribbon Research Labs | 2021 – Present

- ▶ Architected analytics systems processing 3B events/hour; designed 5G VoNR analytics for operator network visibility.
- ▶ Pioneered on-premises LLM deployments (NVIDIA A100/H100/L40S); built RAG chatbot serving 10M+ tokens, \$100K+ savings.
- ▶ Engineered RfX Responder with agentic AI — reduced sales first response from months to days; accuracy 75% – 87%.
- ▶ Built Acumen AI Cluster (5-node K8s, 12 GPUs, 230 GB VRAM) and Acumen AI Platform (FastAPI + Celery + MCP Servers).
- ▶ Fine-tuned LLMs using Unsloth/LoRA/QLoRA/RLHF; deployed Langflow for low-code agentic workflow democratization.
- ▶ AI/ML Steering Committee member; introduced Codeium, Windsurf, Cursor across the org; moderated internal AI/ML forum.

Cisco Systems India Pvt Limited | Software Engineer | 2015 – 2021

- ▶ Architected 5G-NSA solution on MME/PGW for world's first 5G-NSA deployment at T-Mobile.
- ▶ Owned QoS Modeling in Cisco's industry-first Cloud Native Standalone 5G Core (SMF), SMF-UPF and SMF-CHF interfaces.
- ▶ Developed S2a (SaMOG) Wi-Fi Offload gateway features; prototyped MPTCP on DPDK for high-throughput networking.

Samsung Research Institute | Chief Engineer | 2014 – 2015

- ▶ Led TAC team for Jio — world's first greenfield all-IP 4G/LTE network; managed KPIs for 100,000+ eNodeBs.

Nokia Solutions & Networks | Product Architect | 2008 – 2014

- ▶ Architected and delivered world's first commercial 3GPP Femto Gateway (HNbGW) for SFR, France.
- ▶ FCAPS management for 60,000 concurrent Femto APs with 99.999% availability. Led Agile transformation for 120-member team.
- ▶ Pioneered CI/CD before Jenkins: 5,000+ automated test cases in 3-hour window (2013). Prototype of Health Monitor Console predating Prometheus/Grafana.

Motorola India Pvt Limited | Software Engineer | 2004 – 2008

- ▶ Developed Feature Database for P2K Platform; implemented OTA flex file updates; automated systems scaling 6X.

SKILLS & TECHNOLOGIES

AI/ML & Agentic

LLMs, SLMs, RAG, LoRA, QLoRA, RLHF, Unsloth, vLLM, Ollama, LangChain, LangGraph, LlamaIndex, Langfuse, Langflow, n8n, HuggingFace, Arize-Phoenix, Claude Code, Copilot, Windsurf, Cursor

Cloud & Infrastructure

Kubernetes, Docker, FastAPI, Celery, Flask, Milvus, Redis, PostgreSQL, ClickHouse, AWS Bedrock, GCP, Azure

Programming

Python, JavaScript, C/C++, Golang; Full-stack: FastAPI, Flask, React, gRPC, Protobuf, REST APIs, DevOps, Git, OpenTracing

Monitoring & Observability

Prometheus, Grafana, Arize-Phoenix, Langsmith, Langfuse, Blackbox

Telecom & Networking

3G/4G LTE/5G, VoLTE, VoNR, 5G SMF/MME/PGW/UPF, P4/Tofino, eBPF, DPDK, MPTCP, SaMOG, Femto GW, FCAPS

Data Engineering

Hadoop, Parquet, ClickHouse, Milvus, Redis, PostgreSQL; Synthetic data pipelines, fine-tuning dataset generation

EDUCATION & PATENT

M.S. Software Systems — BITS, Pilani

US Patent: US011350319B2

CGPA: 7.26 | 2011

B.E. Information Science — AIT Chickmagluru

76% | 2004

Optimized Quality of Service Enforcement for Handover
Between Different Radio Access Types